

Multi Agentic Workflow for student Attendance Recognition and Real-time Monitoring LMS

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Abstract — *Multi Agentic Workflow for Student Attendance Recognition and Real-time Monitoring LMS* is an AI-powered Learning Management System designed to improve online classroom monitoring by combining Computer Vision and Multi-Agent Intelligence. In many online learning platforms, attendance is recorded only when students join the class, making it difficult for faculty to know whether students remain present, attentive, or are being represented by someone else during the session. The proposed system addresses these challenges by continuously monitoring student participation through live video analysis. It uses OpenCV and MediaPipe to perform face detection, facial landmark extraction, and head-pose estimation for identifying student presence and attention levels. The system includes independent AI agents such as the Attendance Agent, Engagement Agent, and Integrity Agent, which work together to verify attendance, analyze student engagement, and detect proxy attendance or identity changes during live classes. The collected information is securely stored and presented to faculty through an interactive dashboard that provides attendance reports, engagement analytics, and integrity alerts. The system also supports role-based access for students, faculty, department administrators, and administrators, making it suitable for educational institutions. By integrating intelligent monitoring with an LMS, the proposed system improves attendance accuracy, promotes academic integrity, enhances student participation, and provides a smarter and more reliable online learning environment.

Keywords: Learning Management System (LMS), Multi-Agent Systems, Artificial Intelligence (AI), Computer Vision, Student Attendance Recognition, Real-Time Monitoring, Face Detection, Face Recognition, MediaPipe, OpenCV, Student Engagement Analysis, Head Pose Estimation, Identity Verification, Online Learning, Academic Integrity.

I. INTRODUCTION

The rapid growth of digital technologies has transformed the education sector by making online learning more accessible through **Learning Management Systems (LMS)**. Today, universities and educational institutions use LMS platforms to conduct virtual classes, share study materials, manage assignments, and maintain communication between students and faculty. Although these platforms have made online education more convenient, most existing systems still rely on basic attendance methods that only record whether a student has joined the online session. They cannot determine whether the student remains present, attentive, or actively participates throughout the class. One of the major challenges in online education is maintaining **academic integrity** and ensuring genuine student participation. In many cases, students may leave the session after marking attendance, remain distracted during lectures, or allow another person to attend the class on their behalf. Existing Learning Management Systems generally do not provide continuous attendance verification or real-time engagement analysis.

To overcome these limitations, this paper proposes Multi Agentic Workflow for Student Attendance Recognition and Real-time Monitoring LMS, an AI-powered classroom monitoring system that combines Computer Vision and Multi-Agent Intelligence. The system continuously analyzes live video streams using OpenCV and MediaPipe to detect faces, identify facial landmarks, and estimate head pose for monitoring student presence and attentiveness. It introduces three intelligent software agents—Attendance Agent, Engagement Agent, and Integrity Agent—that work together to verify attendance, monitor engagement levels, and detect proxy attendance or identity changes during live online sessions.

The proposed system also provides role-based access for students, faculty members, department administrators, and system administrators through an integrated Learning Management System. All attendance records, engagement reports, and integrity alerts are securely stored and displayed on interactive dashboards, helping educators monitor classroom activities more effectively. By integrating artificial intelligence with modern LMS technologies, the proposed system aims to improve attendance accuracy, increase student engagement, support academic integrity, and create a smarter, more secure, and more interactive online learning environment. This project demonstrates how Computer Vision and Multi-Agent AI can be practically applied to enhance the quality and effectiveness of digital education.

II. SYSTEM ANALYSIS

System analysis is an important phase in the development of any software application, as it helps identify the requirements, limitations, and expected functionalities of the proposed system. The **Multi Agentic Workflow for Student Attendance Recognition and Real-time Monitoring LMS** is designed to improve online education by providing intelligent attendance monitoring, student engagement analysis, and identity verification during live virtual classes. The system combines Computer Vision with Multi-Agent Intelligence to create a secure and efficient learning environment for both students and faculty.

The proposed system mainly involves four types of users: Students, Faculty Members, Department Administrators, and System Administrators. Students can attend live online classes, access learning materials, submit assignments, and view their attendance records. Faculty members can conduct online classes, monitor student participation, and access attendance and engagement reports. Department administrators manage department-related academic activities, while system administrators maintain user accounts, academic information, and overall system performance. By integrating these features into a single Learning Management System, the proposed solution simplifies online classroom management and improves communication between all users.

A. Existing System

The existing Learning Management Systems (LMS) provide online classes, attendance recording, assignment management, and communication between students and faculty. However, they cannot continuously monitor student presence, engagement, or identity during live sessions, which makes it difficult to ensure academic integrity.

Limitations of Existing System

- Basic attendance recording only.
- No engagement monitoring.
- Cannot detect proxy attendance.
- Limited real-time monitoring.

B. Proposed System

The proposed system, **Multi Agentic Workflow for Student Attendance Recognition and Real-time Monitoring LMS**, is an AI-powered Learning Management System that uses Computer Vision and Multi-Agent Intelligence to monitor student attendance and engagement during live online classes. It continuously verifies student presence, detects proxy attendance, and provides real-time reports through an interactive dashboard for faculty and administrators.

Key Features

- Continuous attendance monitoring.
- Student engagement analysis.
- Identity verification.
- Real-time dashboard and reports.
- Multi-Agent AI for intelligent monitoring.

Advantages:

Continuous Attendance Monitoring:

The system continuously verifies whether students remain present throughout the online class instead of recording attendance only at the beginning of the session.

Student Engagement Analysis:

The system monitors facial orientation, head movements, and attention levels to identify whether students are actively participating during live classes.

Identity Verification:

The proposed system detects possible proxy attendance and identity changes, helping educational institutions maintain academic integrity.

Real-Time Dashboard:

Faculty members can access attendance records, engagement reports, and integrity alerts through an interactive dashboard for better classroom monitoring.

C. Functional Requirements

The system includes the following functional requirements:

- User registration and secure login.
- Role-based access for students, faculty, and administrators.
- Online classroom and video conferencing.
- Automatic attendance verification.
- Student engagement monitoring.
- Identity verification and proxy attendance detection.

D. Non-Functional Requirements

The system should also satisfy the following non-functional requirements:

- **Performance:** Fast processing of attendance and real-time monitoring.
- **Security:** Secure login and protection of user data.
- **Scalability:** Supports multiple users and online classes.
- **Reliability:** Accurate attendance and engagement monitoring.

III. METHODOLOGY

A. Architecture

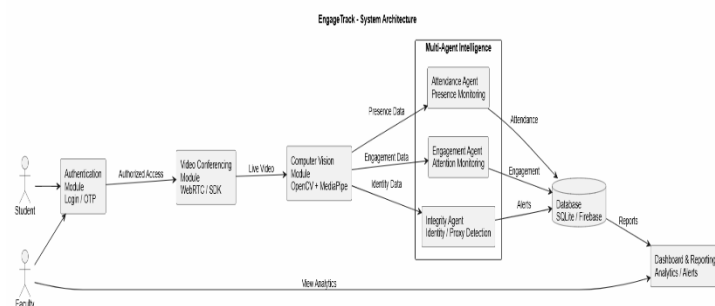


Figure 1: Architecture diagram

The architecture of the Multi Agentic Workflow for Student Attendance Recognition and Real-time Monitoring LMS is designed to provide intelligent attendance and classroom monitoring during live online sessions. The system begins with the Authentication Module, where students and faculty log in using their credentials and OTP verification. After successful authentication, users can access the Video Conferencing Module, which captures the live video stream.

The live video is processed by the Computer Vision Module using OpenCV and MediaPipe to detect faces, facial landmarks, and head movements. The extracted information is then sent to the Multi-Agent Intelligence module, which consists of three independent agents. The Attendance Agent verifies student presence, the Engagement Agent monitors attention levels, and the Integrity Agent detects identity changes or proxy attendance.

The results generated by these agents are stored in the Database, where attendance records, engagement reports, and alerts are maintained. Finally, the Dashboard and Reporting Module displays attendance statistics, engagement analytics, and integrity alerts to faculty members for effective classroom monitoring.

B. Process

I. Modules

1. Authentication Module

This module allows students and faculty to log in securely using their credentials and OTP verification. It ensures that only

authorized users can access the Learning Management System and online classes.

2. Video Conferencing Module

This module provides the live online classroom environment. It captures the student's live video stream, which is used for attendance verification and engagement monitoring.

3. Computer Vision Module

This module processes the live video using OpenCV and MediaPipe. It detects faces, extracts facial landmarks, and tracks head movements for further analysis.

4. Attendance Agent Module

The Attendance Agent continuously verifies whether the student is present during the online session. It automatically records attendance and detects prolonged absence.

5. Engagement Agent Module

This module analyzes facial orientation, eye focus, and head pose to determine the student's attention level during the class.

6. Integrity Agent Module

The Integrity Agent verifies student identity throughout the session and detects proxy attendance or identity changes to support academic integrity.

7. Database Management Module

This module stores user details, attendance records, engagement reports, session logs, and integrity alerts securely in the database.

8. Dashboard and Reporting Module

This module provides attendance reports, engagement analytics, and integrity alerts through an interactive dashboard for faculty and administrators.

which the system identifies their role and provides access to the appropriate dashboard and features.

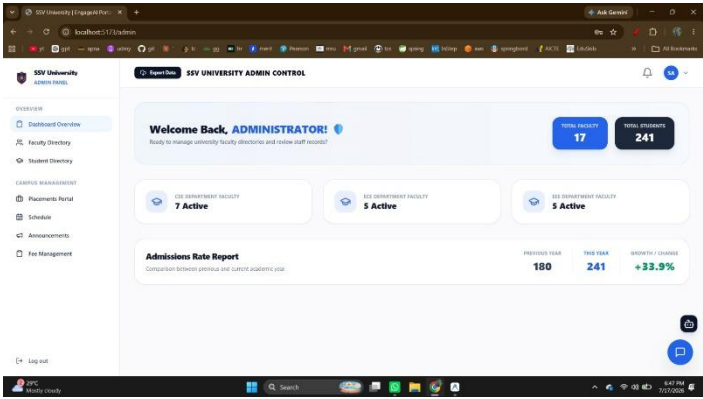


Figure 2: Admin Dashboard

The dashboard also provides access to key modules including the Faculty Directory, Student Directory, Placements Portal, Schedule, Announcements, and Fee Management. It displays an Admissions Rate Report comparing current and previous academic-year admissions, helping administrators track institutional growth. Overall, the dashboard enables efficient management of academic records, faculty and student information, and major campus activities from a single interface.

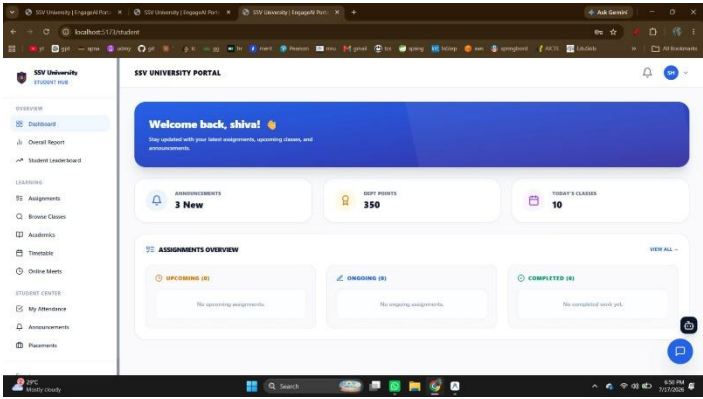


Figure 3: Student Dashboard

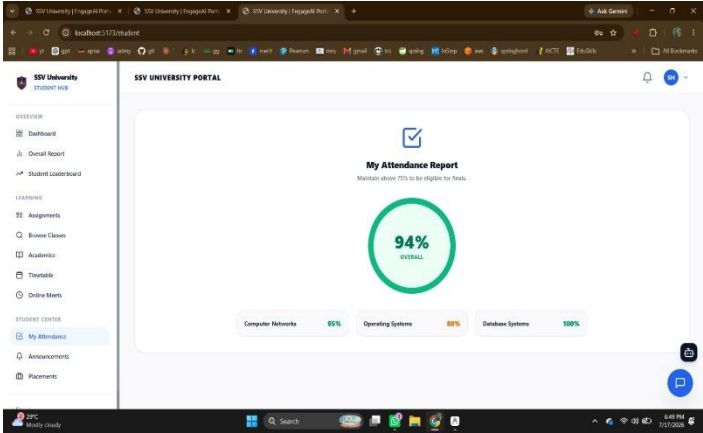


Figure 3.1: Student Attendance tracking

IV. Results

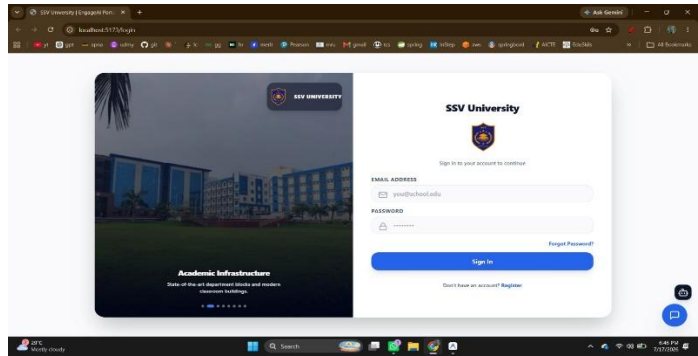


Figure 1: Login Page

The Main Login Page serves as the common entry point for all authorized users of the Multi Agentic Workflow for student Attendance Recognition and Real-time Monitoring LMS system, including Students, Faculty, Department Admins, and Main Admins. Users sign in using their registered email address and password, after

The Student Dashboard provides a centralized interface for students to manage their academic activities. It allows access to assignments, online classes, attendance, timetable, announcements, placements, and reports. The dashboard also displays important updates and helps students track their academic progress from a single platform.

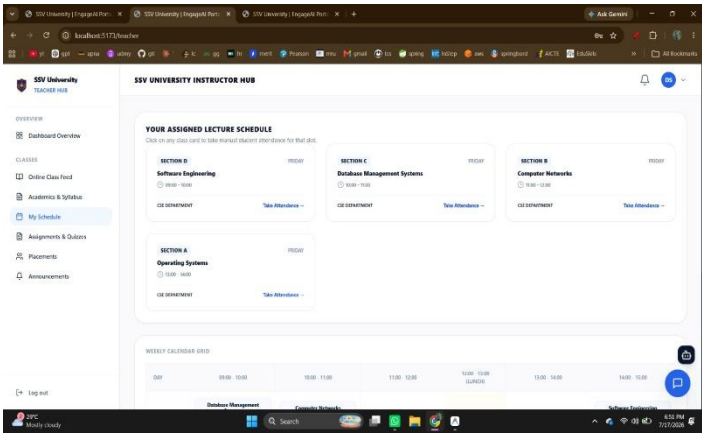


Figure 4: Faculty Dashboard

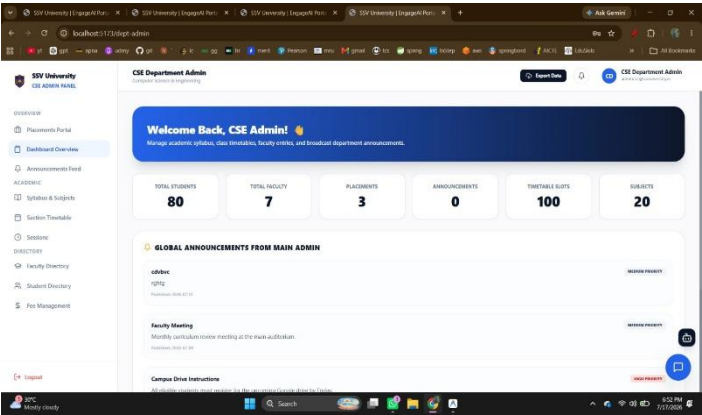


Figure 5: Department Admin Dashboard

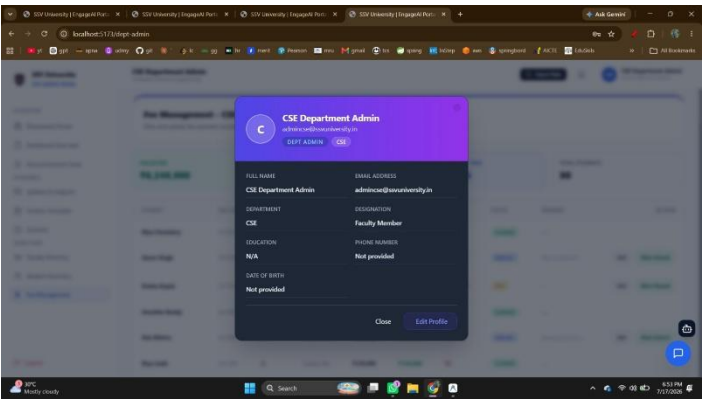


Figure 5.1: Department Admin Dashboard

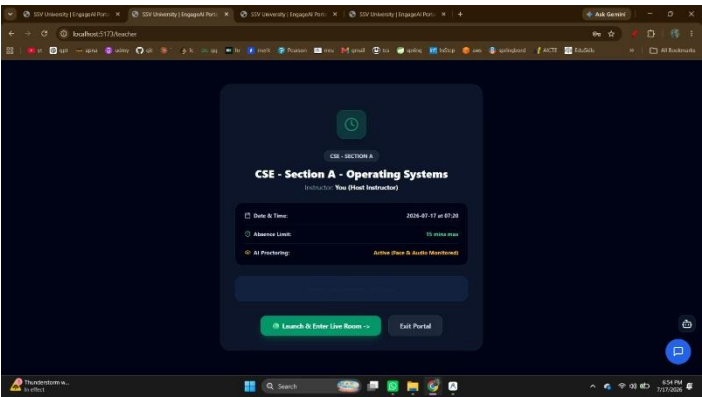


Figure 6: Online Meeting Schedule

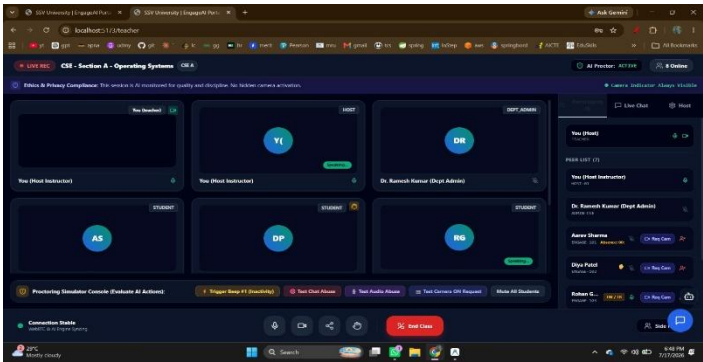


Figure 6.1: Online Video Conference/Meeting

The Video Conference Meeting module provides a live virtual classroom for students and faculty. It supports features such as microphone, camera, screen sharing, chat, and participant management. The system also uses AI-based monitoring to track student attendance, engagement, and identity during online classes while providing real-time alerts and reports to faculty.

V.CONCLUSION

In conclusion, the **Multi Agentic Workflow for Student Attendance Recognition and Real-time Monitoring LMS** is an effective and intelligent solution for improving online classroom management. The system combines **Computer Vision**, **Artificial Intelligence**, and **Multi-Agent Intelligence** to automate attendance verification, monitor student engagement, and detect proxy attendance or identity changes during live online classes. Unlike traditional Learning Management Systems that only record attendance at the beginning of a session, the proposed system continuously monitors student participation throughout the class, providing faculty members with accurate and reliable information. The system integrates multiple modules such as secure user authentication, video conferencing, computer vision processing, attendance monitoring, engagement analysis, identity verification, database management, and dashboard reporting into a single Learning Management System. Faculty members can easily access attendance reports, engagement analytics, and integrity alerts through interactive dashboards, enabling them to make better academic decisions and improve classroom management.

The proposed system is developed using modern technologies such as Python, OpenCV, MediaPipe, FastAPI, React, and SQLite/Firebase, which provide a scalable, reliable, and user-friendly platform. Its modular architecture allows the system to be easily maintained and upgraded as new technologies become available. The role-based access control also ensures that students, faculty members, department administrators, and administrators can securely access the features relevant to their responsibilities.

Overall, the Multi Agentic Workflow for Student Attendance Recognition and Real-time Monitoring LMS demonstrates the practical application of Artificial Intelligence and Computer Vision in online education. It improves student participation, supports academic integrity, enhances the quality of virtual classrooms, and provides a smarter approach to attendance management. The proposed system offers a strong foundation for future intelligent Learning Management Systems and contributes to creating a more interactive, secure, and efficient digital learning environment.

VI.FUTURE SCOPE

The **Multi Agentic Workflow for Student Attendance Recognition and Real-time Monitoring LMS** has significant potential for future improvements and enhancements. In the future, the system can be integrated with advanced Artificial Intelligence and Deep Learning models to improve the accuracy of face recognition, engagement analysis, and identity verification. Emotion recognition can also be added to understand students' learning behavior and provide better insights into classroom participation. The system can be enhanced with speech analysis and gesture recognition to evaluate student interaction more effectively during live classes. AI-based performance prediction and personalized learning recommendations can also be introduced to help students improve their academic performance. Integration with cloud computing services will enable the system to support a larger number of users and online classrooms while ensuring high availability and scalability. Future versions of the system can include a dedicated mobile application for Android and iOS, allowing students and faculty to access classes and monitoring reports from anywhere. Features such as multilingual support, automated attendance notifications, smart report generation, and integration with popular Learning Management Systems like **Moodle** and **Google Classroom** can further improve usability. Security can be strengthened by implementing biometric authentication, advanced encryption techniques, and secure cloud storage to protect user data and attendance records. The system can also be expanded to support hybrid classrooms, where both online and offline students are monitored simultaneously. Overall, these enhancements will make the proposed system more intelligent, secure, scalable, and suitable for modern digital education environments.

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